

Call: HORIZON-MSCA-2024-PF-01
(MSCA Postdoctoral Fellowships 2024)

Topic: HORIZON-MSCA-2024-PF-01-01

Type of Action: HORIZON-TMA-MSCA-PF-EF
(HORIZON TMA MSCA Postdoctoral Fellowships - European Fellowships)

Proposal number: 101205639

Proposal acronym: COGITO

Type of Model Grant Agreement: HORIZON Unit Grant

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1 - General information

Fields marked * are mandatory to fill.

Topic	HORIZON-MSCA-2024-PF-01-01	Type of Action	HORIZON-TMA-MSCA-PF-EF
Call	HORIZON-MSCA-2024-PF-01	Type of Model Grant Agreement	HORIZON-AG-UN

Acronym COGITO

Proposal title

Cognition and Coding: How Computational Thinking and Programming Learning is Impacted by Digital Working Memory Intervention in Primary School Children.

Note that for technical reasons, the following characters are not accepted in the Proposal Title and will be removed: < > " &

Scientific Area

SOC - Social Sciences and Humanities (SOC)

Please select up to 5 descriptors (and at least 3) that best characterise the subject of your proposal, in descending order of relevance.

Descriptor 1

Cognition (e.g. learning, memory, emotions, speech)

Descriptor 2

Educational psychology

Descriptor 3

Pedagogy

Descriptor 4

Multimedia

Free keywords

Working Memory, Programming, Early Intervention

Please choose the scientific area and descriptors carefully, and in order of importance, since this will guide the REA in the selection of experts for proposal evaluation and the allocation of proposals to experts.

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Abstract *

Digital competency is highly sought after, but the cognitive and motivational aspects of computational thinking (CT) and learning to program are still largely unknown. Here we will investigate: 1. The cognitive aspects important for computational thinking and programming. 2. The motivational and personality aspects associated with computational thinking. 3. Whether mathematical and cognitive training, prior to learning to program, can promote programming learning in young children.

Two yearly cohorts of 150 children will train for 8 weeks using the free application "Vektor" and, subsequently, they and 150 controls will learn programming with ScrathJr in a 5 week course. All delivered by their regular teachers. They will all then be evaluated one-on-one by a team of 3 researchers to measure how advanced their coding knowledge and computational thinking skills have become.

Our increasingly technical societies demand and are expected to continue to demand CT skills and mathematics as they are both predictors of STEM field performance. Swedish schools have incorporated CT as a required part of their curriculum, but no consensus as to what to teach, or how best to do so has been provided. Since the idea is to improve broadly applicable cognitive skills, evaluating if cognitive training in combination with early exposure to programming can lead to transferable skills is a promising prospect given the close relationships discovered between CT, programming, working memory and math.

Remaining characters 500

Has this proposal (or a very similar one) been submitted in the past 2 years in response to a call for proposals under any EU programme, including the current call?

☐ Yes ☒ No

Please give the proposal reference or contract number.

Previously submitted proposals should be with either 6 or 9 digits.

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Declarations

Field(s) marked * are mandatory to fill.

1) We declare to have the explicit consent of all applicants on their participation and on the content of this proposal. * ☒

2) We confirm that the information contained in this proposal is correct and complete and that none of the project activities have started before the proposal was submitted (unless explicitly authorised in the call conditions). * ☒

3) We declare:

- to be fully compliant with the eligibility criteria set out in the call
- not to be subject to any exclusion grounds under the [EU Financial Regulation 2018/1046](#)
- to have the financial and operational capacity to carry out the proposed project. *

☒

4) We acknowledge that all communication will be made through the Funding & Tenders Portal electronic exchange system and that access and use of this system is subject to the [Funding & Tenders Portal Terms and Conditions](#). * ☒

5) We have read, understood and accepted the [Funding & Tenders Portal Terms & Conditions](#) and [Privacy Statement](#) that set out the conditions of use of the Portal and the scope, purposes, retention periods, etc. for the processing of personal data of all data subjects whose data we communicate for the purpose of the application, evaluation, award and subsequent management of our grant, prizes and contracts (including financial transactions and audits). * ☒

6) We declare that the proposal complies with ethical principles (including the highest standards of research integrity as set out in the [ALLEA European Code of Conduct for Research Integrity](#), as well as applicable international and national law, including the Charter of Fundamental Rights of the European Union and the European Convention on Human Rights and its Supplementary Protocols. [Appropriate procedures, policies and structures](#) are in place to foster responsible research practices, to prevent questionable research practices and research misconduct, and to handle allegations of breaches of the principles and standards in the Code of Conduct. * ☒

7) We declare that the proposal has an exclusive focus on civil applications (activities intended to be used in military application or aiming to serve military purposes cannot be funded). If the project involves dual-use items in the sense of [Regulation 2021/821](#), or other items for which authorisation is required, we confirm that we will comply with the applicable regulatory framework (e.g. obtain export/import licences before these items are used). * ☒

8) We confirm that the activities proposed do not

- aim at human cloning for reproductive purposes;
- intend to modify the genetic heritage of human beings which could make such changes heritable (with the exception of research relating to cancer treatment of the gonads, which may be financed), or
- intend to create human embryos solely for the purpose of research or for the purpose of stem cell procurement, including by means of somatic cell nuclear transfer.
- lead to the destruction of human embryos (for example, for obtaining stem cells)

These activities are excluded from funding. * ☒

9) We confirm that for activities carried out outside the Union, the same activities would have been allowed in at least one EU Member State. * ☒

The coordinator is only responsible for the information relating to their own organisation. Each applicant remains responsible for the information declared for their organisation. If the proposal is retained for EU funding, they will all be required to sign a declaration of honour.

False statements or incorrect information may lead to administrative sanctions under the EU Financial Regulation.

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2 - Participants

List of participating organisations

#	Participating Organisation Legal Name	Country	Role	Action
1	KAROLINSKA INSTITUTET	Sweden	Coordinator	

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Short name KI

Organisation data

PIC	Legal name
999978530	KAROLINSKA INSTITUTET

Short name: KI

Address

Street	Nobels Vag 5
Town	STOCKHOLM
Postcode	17177
Country	Sweden
Webpage	www.ki.se

Specific Legal Statuses

Legal person	yes	Academic Sector	yes
Public body	yes		
Non-profit	yes		
International organisation	no		
Secondary or Higher education establishment	yes		
Research organisation	no		

SME Data

Based on the below details from the Participant Registry the organisation is unknown (small- and medium-sized enterprise) for the call.

SME self-declared status	unknown
SME self-assessment	unknown
SME validation	unknown

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Short name **KI**

Departments carrying out the proposed work

Department 1

Department name	Department of Neuroscience	☐ not applicable
	☐ Same as proposing organisation's address	
Street	Solnavägen 9	
Town	Stockholm	
Postcode	17165	
Country	Sweden	

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Supervisor

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title **Prof.**

Gender ☐ Woman ☒ Man ☐ Non Binary

First name* **Torkel**

Last name* **Klingberg**

E-Mail* **torkel.klingberg@ki.se**

Position in org. **Professor**

Department **Department of Neuroscience**

☐ Same as organisation name

☐ Same as proposing organisation's address

Street **Solnavägen 9**

Town **Stockholm**

Post code **17165**

Country **Sweden**

Website **<https://www.klingberglab.se/>**

Phone **+46852486359**

Phone 2 **+xxx xxxxxxxxxx**

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Short name **KI**

Researcher

The name and e-mail of the Researcher and Supervisor are read-only in the administrative form, only additional details can be edited here. To give access rights and contact details of contact persons, please go back to Participant's page of the submission wizard and save the changes.

Last Name*	Lidström	Last Name at Birth	Lidström
First Name(s)*	Andreas	Gender*	☐ Woman ☒ Man ☐ Non binary
Title	Dr	Country of residence*	Sweden
Nationality*	Sweden	Nationality 2	
Date of Birth (DD/MM/YYYY)	17/03/1989	Country of Birth*	Sweden
		Place of Birth	Södertälje

Contact address

Current organisation name	Karolinska Institutet		
Current Department/Faculty/Institute/Laboratory name	Klingberg Lab, Department of Neuroscience		
	☐ Same as organisation address		
Street	Solnavägen 9		
Postcode/Cedex	17165	Town	Stockholm
Phone	+46739977651	Country	Sweden
Phone2 / Mobile	+xxx xxxxxxxxx		
E-Mail*	antecave@gmail.com		
ORCID	0009-0008-7372-8541		
Researcher ID			
	The maximum length of the identifier is 11 characters (ZZZ-9999-2010) and the minimum length is 9 characters (A-1001-2010).		
Other ID	Please enter the type of ID here		Please enter the identifier number here

Qualifications

Doctorate Date of (expected) award	Select the exact date (DD/MM/YYYY)	21/07/2022
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Place of activity/place of residence (previous 5 years - most recent one first)

Indicate the period(s) and the country/countries in which you have legally resided and/or had your main activity (work, studies, etc) during the last 5 years up until the deadline for the submission of the proposal.

Please fill in this section without gaps. Short stays (as defined in the Guide for Applicants) shall not be listed in this box.

Period from	Period to	Duration (days)	Country
04/12/2023	11/09/2024	283	Sweden
14/08/2018	03/12/2023	1938	United Kingdom
Total		2221	

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Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☐
Other	☐
If yes, please specify: (Maximum number of characters allowed: 50)	

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List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Software	<i>"Vektor", is an online Working-Memory cognitive training application for children with automatic progress monitoring where data is saved to a secure database. Designed and provided by Torkel Klingberg lab at Karolinska Institutet, this software is a core resource needed for the proposed project.</i>
Publication	<i>Judd, N.& Klingberg, T. et.al. 2020. Cognitive and brain development is independently influenced by socioeconomic status and polygenic scores for educational attainment. Proceedings of the National Academy for Sciences</i>
Publication	<i>Nemmi F, Helander E, Helenius O, Almeida R, Hassler M, Räsänen P, Klingberg T. (2016) Behavior and neuroimaging at baseline predict individual response to combined mathematical and working memory training in children. Developmental Cognitive Neuroscience 20, 43-51 (2016).</i>
Publication	<i>Constantinidis, C. & Klingberg, T. The neuroscience of working memory capacity and training. Nature Reviews Neuroscience (2016).</i>
Publication	<i>Nemmi F, Schel MA, Klingberg T. (2018) Connectivity of the Human Number Form Area Reveals Development of a Cortical Network for Mathematics. Front Hum Neurosci. Nov 26;12:465.</i>

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
Project HOPE	<i>The interdisciplinary research project HOPE, "Hands-On in computer science-Education", with Professor Sara Bengtsson (department of clinical neuroscience at Karolinska Institutet) & Professor Anna Eckerdal (department of computer science education at Uppsala University), provided the theoretical groundwork from which the current proposal is based, where we mapped the core cognitive machinery of programming learning in young adults by means of fMRI and investigated the role of confidence factors.</i>
ScratchJr Pilot Project	<i>I planned and executed a successful pilot of the first 5 days of the 'Coding as Language' (CAL) ScratchJr Programming course and its evaluation-material ('Tech-Check' test of computational thinking, and the CSA 'coding stages assessment' of programming attainment) that will form a major part of the proposed MRCA project in one school.</i>
Trippel Helix Conference Presentation	<i>Engaged with key stakeholders from the private and public sector by spreading awareness of project HOPE via participation in a discussion panel at the Trippel Helix Conference on Computational Thinking and Digital Competences in Primary and Secondary Education, Stockholm, September 8, 2017. Organized by Swedsoft, a non-profit organization working to promote competitive software engineering in Sweden.</i>

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Biomedicum	<i>Laboratory and Office complex on the campus of Karolinska Institutet housing Klingberg Lab.</i>

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Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☒ Yes

☐ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

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3 - Budget

Is the Researcher eligible for family allowance?* ☐ Yes ☒ No

Duration of fellowship *	Country in which fellowship will take place
24	Sweden

	Country Coefficient	Number of Months	Contributions for recruited researchers			Institutional contributions		Total
			Living Allowance	Mobility Allowance	Family Allowance	Research, training and networking costs	Management and indirect costs	
Fellowship	1.25	24	179700.00	17040.00	0.00	24000.00	15600.00	236340.00
Total			179700.00	17040.00	0.00	24000.00	15600.00	236340.00

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4 - Ethics & security

Ethics Issues Table

1. Human Embryonic Stem Cells and Human Embryos		Page
Does this activity involve Human Embryonic Stem Cells (hESCs)?	☐ Yes ☒ No	
Does this activity involve the use of human embryos?	☐ Yes ☒ No	
2. Humans		Page
Does this activity involve human participants?	☒ Yes ☐ No	17
Are they volunteers for non medical studies (e.g. social or human sciences research)?	☒ Yes ☐ No	17
Are they healthy volunteers for medical studies?	☐ Yes ☒ No	
Are they patients for medical studies?	☐ Yes ☒ No	
Are they potentially vulnerable individuals or groups?	☐ Yes ☒ No	
Are they children/minors?	☒ Yes ☐ No	17
Are they other persons unable to give informed consent?	☐ Yes ☒ No	
Does this activity involve interventions (physical also including imaging technology, behavioural treatments, etc.) on the study participants?	☒ Yes ☐ No	17
Does it involve invasive techniques?	☐ Yes ☒ No	
Does it involve collection of biological samples?	☐ Yes ☒ No	
Does this activity involve conducting a clinical study as defined by the Clinical Trial Regulation (EU 536/2014) ? (using pharmaceuticals, biologicals, radiopharmaceuticals, or advanced therapy medicinal products)	☐ Yes ☒ No	
3. Human Cells / Tissues (not covered by section 1)		Page
Does this activity involve the use of human cells or tissues?	☐ Yes ☒ No	
4. Personal Data		Page
Does this activity involve processing of personal data?	☒ Yes ☐ No	17
Does it involve the processing of special categories of personal data (e.g.: genetic, biometric and health data, sexual lifestyle, ethnicity, political opinion, religious or philosophical beliefs)?	☐ Yes ☒ No	
Does it involve profiling, systematic monitoring of individuals, or processing of large scale of special categories of data or intrusive methods of data processing (such as, surveillance, geolocation tracking etc.)?	☐ Yes ☒ No	
Does this activity involve further processing of previously collected personal data (including use of preexisting data sets or sources, merging existing data sets)?	☐ Yes ☒ No	
Is it planned to export personal data from the EU to non-EU countries?	☐ Yes ☒ No	
Is it planned to import personal data from non-EU countries into the EU or from a non-EU country to another non-EU country?	☐ Yes ☒ No	
Does this activity involve the processing of personal data related to criminal convictions or offences?	☐ Yes ☒ No	
5. Animals		Page
Does this activity involve animals?	☐ Yes ☒ No	

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6. Non-EU Countries	Page
Will some of the activities be carried out in non-EU countries?	☐ Yes ☒ No
In case non-EU countries are involved, do the activities undertaken in these countries raise potential ethics issues?	☐ Yes ☒ No
It is planned to use local resources (e.g. animal and/or human tissue samples, genetic material, live animals, human remains, materials of historical value, endangered fauna or flora samples, etc.)?	☐ Yes ☒ No
Is it planned to import any material (other than data) from non-EU countries into the EU or from a non-EU country to another non-EU country? For data imports, see section 4.	☐ Yes ☒ No
Is it planned to export any material (other than data) from the EU to non-EU countries? For data exports, see section 4.	☐ Yes ☒ No
Does this activity involve low and/or lower middle income countries , (if yes, detail the benefit-sharing actions planned in the self-assessment)	☐ Yes ☒ No
Could the situation in the country put the individuals taking part in the activity at risk?	☐ Yes ☒ No
7. Environment, Health and Safety	Page
Does this activity involve the use of substances or processes that may cause harm to the environment, to animals or plants.(during the implementation of the activity or further to the use of the results, as a possible impact) ?	☐ Yes ☒ No
Does this activity deal with endangered fauna and/or flora / protected areas?	☐ Yes ☒ No
Does this activity involve the use of substances or processes that may cause harm to humans, including those performing the activity.(during the implementation of the activity or further to the use of the results, as a possible impact) ?	☐ Yes ☒ No
8. Artificial Intelligence	Page
Does this activity involve the development, deployment and/or use of Artificial Intelligence-based systems?	☐ Yes ☒ No
9. Other Ethics Issues	Page
Are there any other ethics issues that should be taken into consideration?	☐ Yes ☒ No

I confirm that I have taken into account all ethics issues above and that, if any ethics issues apply, I will complete the ethics self-assessment as described in the guidelines [How to Complete your Ethics Self-Assessment](#)



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Ethics Self-Assessment

Ethical dimension of the objectives, methodology and likely impact

Objectives and Justification for involving children:

I will study transfer effects from WM training to Programming learning (Obj.1), and how non-cognitive personality traits affect this (Obj.2) in primary school age children, since WM training has shown the most positive impact in this age range, and it is also an understudied demographic when it comes to programming learning and the interplay between cognitive and non-cognitive processes of computational thinking (Obj.3).

Methodology:

Non-clinical controlled intervention study where half of the participants will receive the WM training via a custom training app, and the other half will not. The use of children is necessary, being the specific target group of my research objectives. Personal data in the form of personality measures and test results on computational thinking test will be collected by the teachers in whole class, and a final examination of programming aptitude will be performed one-on-one, in person or via Zoom, by me. For a description of the psychological instruments used, and planned analyses see part B26 section 6 (page 17).

Potential impact:

No likely adverse impacts expected. A data security breach could spread personal information in the form of test results linked with participant names, but I judge the potential for misuse of any such data as low. I will not collect personal ID numbers making direct identification non-trivial, and since I am not collecting data falling into any specially protected categories, no social stigma is to be expected in case of a breach. All results will be presented in aggregate, without personal identifiability, and no individual results will be presented or shared with participants.

Recruitment, inclusion and exclusion criteria and informed consent procedures:

I will recruit students by contacting teachers familiar with the WM training app and assigning them randomly to treatment, or control conditions. Study information and consent sheets will be emailed to each teachers for dissemination to the parents of their class for written consent of both guardians with instructions to also guarantee the verbal ascent to participate of their child. The whole class will participate in the training and programming, but I will exclude participants from my analysis if they fail to reach 20 days of training in the training group.

Unexpected findings policy:

I will not collect any data that can result in unexpected findings.

Details on the age range:

I will recruit first and second grade students (7-8 years old) based on previous findings of transfer effects of WM training to mathematics in this age group (Berger et.al. 2020), and because the programming course material I will use is designed for first grade students (Bers et.al. 2023).

Procedures to ensure the welfare of the children:

Informed consent will be sought and right to withdraw made clear. No intrusive procedures will be performed, and all activities will be framed as part of regular education and administered by their regular teachers. In the final one-on one exam I will follow the written guidelines for the course evaluation material by Berger et.al. and establish good rapport with the children before starting. This is an intervention study, but the intervention is not in any way harmful or stressful, and is designed to be fun and engaging for the students. Withholding the WM intervention from the control group is justified since this is a non-clinical setting and it is provided as a research tool to participating schools only and not part of regular curricula in Swedish schools.

Berger, E. M., Fehr, E., Hermes, H., Schunk, D., & Winkel, K. (2020). The impact of working memory training on children's cognitive and noncognitive skills. NHH Dept. of Economics Discussion Paper, (09).

Bers, M. U., Blake-West, J., Kapoor, M. G., Levinson, T., Relkin, E., Unahalekhaka, A., & Yang, Z. (2023). Coding as another language: Research-based curriculum for early childhood computer science. Early Childhood Research Quarterly, 64, 394-404.

Remaining characters

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Compliance with ethical principles and relevant legislations

To ensure compliance with Swedish and EU regulations, ethics approval has been submitted to Etikprövningsmyndigheten, the Swedish ethics board, under case number 2024-04879-01, answer expected by end of September.

Copies of the ethics approval will be made available upon request, together with my Informed consent form and information sheet.

I have read the the European charter for researchers, and I will follow it.

I will also stick to the FAIR principles of data management if awarded.

My planned data falls under the GDPR definition of personal data and will be collected and processed by me. All data will be 'Pseudonymised', with the keys stored in the lab safe of Klingberg lab at the department of neuroscience at Karolinska Institutet.

I will detail my data protection efforts in my data management plan deliverable, with support from the Karolinska Institute Research Data Office (rdo@ki.se).

Data minimisation statement:
I will collect no variables not part of my research objectives. Training days for inclusion. WM performance and programming level to asses objective 1. Need For Cognition, Self-Esteem and big5 personality inventory to assess objective 2. Computational thinking score and subscores of the WM training to asses objective 3.

Remaining characters

3739

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Security issues table

1. EU Classified Information (EUCI) ²		Page
Does this activity involve information and/or materials requiring protection against unauthorised disclosure (EUCI)?	☐ Yes ☒ No	
Does this activity involve non-EU countries which need to have access to EUCI?	☐ Yes ☒ No	
2. Misuse		Page
Does this activity have the potential for misuse of results?	☐ Yes ☒ No	
3. Other Security Issues		Page
Does this activity involve information and/or materials subject to national security restrictions? If yes, please specify: (Maximum number of characters allowed: 1000)	☐ Yes ☒ No	
Are there any other security issues that should be taken into consideration? If yes, please specify: (Maximum number of characters allowed: 1000)	☐ Yes ☒ No	

Security self-assessment

No security issues identified.

Remaining characters 4970

²According to the Commission Decision (EU, Euratom) 2015/444 of 13 March 2015 on the security rules for protecting EU classified information, “European Union classified information (EUCI) means any information or material designated by an EU security classification, the unauthorised disclosure of which could cause varying degrees of prejudice to the interests of the European Union or of one or more of the Member States”.

³Classified background information is information that is already classified by a country and/or international organisation and/or the EU and is going to be used by the project. In this case, the project must have in advance the authorisation from the originator of the classified information, which is the entity (EU institution, EU Member State, third state or international organisation) under whose authority the classified information has been generated.

⁴EU classified foreground information is information (documents/deliverables/materials) planned to be generated by the project and that needs to be protected from unauthorised disclosure. The originator of the EUCI generated by the project is the European Commission.

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5 - Other questions

Information on the Researcher (future fellow)

1. Were you in the last 3 years in compulsory national service? ☐ Yes ☒ No

2. Did you spend time, in the last 3 years, on procedures for obtaining refugee status (according to the 1951 Geneva Refugee Convention and the 1967 Protocol) in a Member State or Associated Country to Horizon Europe? ☐ Yes ☒ No

3. Are you a national of a Member State or Associated Country? ☒ Yes ☐ No

Country

Other Questions

4. Are the research and training activities proposed addressing topics related to nuclear sciences and technology, as outlined in the Guide for applicants for this call and the Euratom Research and Training Programme (ERTP)? ** ☐ Yes ☒ No

Answer "Yes" ONLY IF all three conditions below are fulfilled:

- The proposal's research area is covered by the ERTP
- The host organisation (and, if applicable, the Associated Partner for the additional Placement period) is/are established in a Member State or Associated Country to the ERTP
- The researcher is a national or a long-term resident of a Member State or Associated Country to the ERTP

5. For communication purposes only, the European Commission REA asks for permission to publish the name of the researcher (future fellow) should the proposal be retained for funding. Does the researcher (future fellow) give this permission? ☒ Yes ☐ No

6. Some national and regional public research funding authorities run schemes to fund MSCA applicants that score highly in the MSCA evaluation but which cannot be funded by the MSCA due to their limited budget. In case this proposal could not be selected for funding by the MSCA, do the researcher and supervisor consent to the European Commission disclosing to such authorities the results of its evaluation (score and ranking range) together with their names and contact details, non-confidential proposal title and abstract, proposal acronym, and host organisation? ☒ Yes ☐ No

Part B-1**1. Excellence #@REL-EVA-RE@#****1.1 *Quality and pertinence of the project's research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art)***

R&I challenge in educational psychology I aim to address in this proposal: Digital competency is becoming increasingly important for individuals and society at large. Schools are now required to include both the fostering of digital competency and computer programming in the curriculum, but we still know very little about how this is best accomplished. A key aspect is a better understanding of “computational thinking” (CT) - what cognitive and motivational aspects that are important for learning to program, especially in young children, and how best to prepare them for the learning of programming concepts. Project COGITO, (Cognition & Coding) will attempt to answer these questions by means of the following research objectives:

[Objective 1]: I will investigate if primary school children trained with the digital Working Memory (WM) training program ‘Vektor’ will show increased learning of programming concepts and CT skills after completing the ‘Coding as Language’ (CAL) curriculum for ScratchJr programming when compared to children without the WM training.

[Objective 2]: The impact of self-esteem, Need for Cognition (NFC) and the big-5 personality dimensions on learning outcomes will be investigated.

[Objective 3]: The constituent cognitive components of CT will be explored by looking at how the comparative impacts of subcomponents of the Vektor training data (mathematics, visuo-spatial working memory (VSWM), and non-verbal reasoning (NVR)) impacts performance on the ‘TechCheck’ computational thinking test.

Pertinence and State-of-the-art: In 2017, the report ‘Digital skills in the EU labour Market’ was published, where it was identified that 40% of the EU workforce had “little to no digital skill” while predicting a future where nearly all jobs will require digital skills (Kiss, 2017), highlighting the importance of teaching computer literacy to future generations. The same year, **Swedish schools were instructed to strengthen the “digital competency” of children, including by introducing programming as a “clear part of several subjects”,** primarily in technical and mathematical courses (Regeringskansliet 2017-03-09). However, how young children should best learn to program is still an understudied area. We still do not know the key cognitive aspects required or the optimal learning strategies, which could depend on different personality traits, or whether children can be better prepared for learning programming by prior mathematical or cognitive training. This mismatch between what is asked of the school system and the state of the current knowledge in this area will be summarised below.

The term computational thinking itself is very broad and has been applied to a wide array of cognitive tasks like programming-specific processes on one hand, to more general problem-solving skills and ways of breaking down problems into sub-tasks, like in algorithmic thinking, on the other (Tang et al., 2020). As already discussed above, the Swedish curriculum, in line with many other EU nations and the UK, instructs teachers to introduce computational thinking into their teaching, but what that should look like in practice is up to the individual schools or teachers, and given how widely defined the term is in the scientific literature, the extent to which computational thinking actually is integrated into teaching, and how effective that teaching is, will vary significantly between regions and individual schools, and it is not yet clear if any meaningful effects have been achieved at the national level from this new directive, even though many studies have shown promising results from computational thinking interventions (Rodrigues, R. S. et al., 2016; Lei, H. et al., 2020; Ye, J. et al., 2022).

One of the more interesting but also poorly understood aspects of teaching programming is characterising which skills and cognitive processes are required, and secondly, which learning strategies are optimal. **A recent systematic review of introductory programming highlighted the need for quantitative studies on learning strategies, and called for investigations into the role of self-efficacy processes such as confidence, stress, and motivation** (Luxton-Reilly et al., 2018). Partly to address this, the most recent of the very few neuroimaging studies of programming to date, conducted by myself, and the only one to focus on learning specifically (Lidström, 2022 Thesis, Manuscript submitted for review), found that **the best predictors of learning and long-term retention was mathematics, confidence, and need for cognition (NFC).** The study also highlighted the importance of VSWM-regions in addition to language-processing for programming task performance. **I was able to show that the VSWM-**

areas in the brain are both critically important when solving programming tasks, and also highly plastic, laying down behaviorally significant connectivity changes after only a few hours of education. **This visuospatial network links WM, mathematics and programming skills cognitively, and we hypothesise that this forms the basic neural substrate for all computational thinking skills.**

Numerous previous studies have also shown that early mathematical ability is a fundamental precursor of successful future learning in science, technology, engineering and mathematics and a good general indicator of broader school performance (Duncan et al., 2007), and there are numerous studies linking mathematical ability to spatial rotation (Wai et al., 2009; Mix et al., 2016; Hawes, 2020) , visuo-spatial working memory (Gathercole et al., 2003; Peng et al., 2016) and non-verbal reasoning (Geary, 2011). **In a landmark study performed by my PI (Judd and Klingberg, 2021) more than 17.000 children were randomised to different regimes of maths learning combined with various amounts of training on cognitive tasks requiring spatial processing, VSWM or NVR.** It was shown that the type of cognitive content matters for how much children improved on mathematical tests that were administered continuously over 8 weeks of training. This study demonstrates that cognitive training of WM transfers to improved mathematical learning. Secondly, it was found that both NVR and VSWM were more effective than pure spatial training. Thirdly, post-hoc analysis suggested that children who were low performers benefited more from NVR, while the opposite trend was found for VSWM, highlighting the **importance of individual differences for optimal interventions.**

Since my previous study has shown the importance of non-cognitive personality traits like confidence and NFC to successful programming learning, it seems only natural to further characterise the relationships between programming learning, personality, and computational thinking skills more thoroughly. Computational thinking has been shown to correlate weakly with openness, conscientiousness and emotional stability, (Román-González et al., 2016; Zhang, S., & Wong, G. K. 2024) and the authors conclude that taking individual attitudes and preferences into account, particularly affection for the subject, can have profound implications for teaching computational thinking skills. **How the training of basic cognitive skills interacts with the non-cognitive personality aspects however remains largely unknown.** To add to this, the majority of previous studies aimed at these questions have used older participants, consequently, **we also know much less about these dynamics in young children.**

The broader field of education is progressively coming around to the idea of embracing what has been termed ‘mind brain and education’, that is the fusion of cognitive neuroscience and educational science. This includes the much narrower field of educational psychology of programming and computational thinking, where only seven papers had been published attempting to investigate the cognitive machinery responsible for various aspects of programming using fMRI or EEG before my own PhD thesis work in this area, and all of those previous studies were either pioneering, but methodologically limited or focused specifically on debugging and code-review, (An area of interest for rating experienced programmers in higher education or industry), rather than on more fundamental cognitive mapping.

Taken together, this proposed project will **go beyond the state of the art** by applying my findings from the first thorough exploration of the cognitive machinery and personal factors needed for successful programming learning, and exploring the potential benefits of WM training as suggested by the overlap in brain regions between programming tasks and WM tasks. Furthermore, I will test this hypothesis in the **under-studied population of young children programming**, and the study of non-cognitive personality traits for this purpose is also novel for this target group. This is also an **ambitious** project, as the effect sizes of previous studies of transfer effects of WM training indicate that a sample size of 600 children might be needed to reach statistical significance, and coordinating with multiple schools and teachers will be required (Berger, E. M. et.al. 2020, 2024).

1.2 Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality of open science practices).

Methodology: The experimental part of my project comprises three work packages. **Preparation and Administration (WP1)** contains tasks needed at project start and before each cohort, described next. I will collect data from **two Cohorts (WP2 & WP3)**, one each year, each corresponding to a deliverable containing half of my

total planned data set (**D2.1** & **D3.1**). **WP1, Preparation and Administration** contains three main tasks: **Ethics & information meetings (Task1.1)** is to have each teacher send out info and consent-forms to the guardians of each student to be archived by me (**M1.1**). I will also host information meetings over Zoom to instruct teachers on how to perform each step of this study before each cohort can start (**M1.2** for cohort 1). **Recruiting(Task1.2)** assistants for the CSA (detailed below)(**M1.3** for cohort 1) and teachers for the next cohort if needed(continuous). 1st-2nd year school children (aged 6-8) (a total of n=600), will be recruited class-wise from primary schools in Sweden by emailing teachers from the database of users of the Vektor software. **Translating(Task1.3)** the psychological instruments and the course material (detailed below) into Swedish, which becomes a deliverable resource (**D1.1**). **WP2** and **WP3** represent the **two cohorts** of data collection (300 students each) and are identical in procedure. After informed consent has been received from all legal guardians, In the spring, each half of the cohort (150) will be assigned to either training with the Vektor **WM training** app (15min/day for 8 weeks) or **no training**. The following autumn term the teachers will administer to the whole class: The Need For Cognition scale, The Rosenbergs Self-Esteem scale, The Pictorial Personality Traits Questionnaire for Children (PPTQ-C), and the TechCheck test of CT skills as a pre-test (**O2:Non-Cognitive Learning Interactions**). They will then work through the CAL ScratchJr programming course (23x45-minute lessons, one per day). After completing the course, TechCheck will be used as a post-test for CT skills and compared with the pre-test (**O3:Computational Thinking Training**). The final data collection, and the most intensive part of this experiment is administering the CSA evaluation of programming learning (My final outcome variable, the basis for **O1:WM Transfer to Programming**). The CSA is designed to be administered individually to each participant and takes up to 45 minutes to complete. This will be done in person or over Zoom. The 300 hours required for each cohort will be split up into two groups of 150 by staggering the start of the CAL course to, allowing time for me, together with two students or research assistants to complete each group of 150 students in under two weeks. (7 students per day for 7 days x 3 evaluators = 150), meaning the whole cohort of 300 can be evaluated in one month of focused work (**Task 2.1-2 for WP2, and Task 3.1-2 for WP3**, resulting in **D2.1** preliminary- and **D3.1** final data sets.).

Interdisciplinary Approaches: This project is a natural continuation of my work on the neural underpinnings of learning programming, and the result of combining knowledge gained from that brain imaging experiment and directly applying the results practically in a classroom setting. I will be using course material developed by specialists in the field of education of programming for children, and combining this with psychologically validated non-cognitive personality measurements from the field of psychometrics and I will be using digital tools developed to study cognitive development. The idea is to fully embody the idea of “mind, brain and education” by including all three aspects. By planning active two-way communication with the involved teachers, I also hope to gain new ideas and insights whilst also promoting greater interest in further research participation from education professionals, thereby also gaining further interdisciplinary skills.

Sex & Gender Dimensions: It is well known that there are consistent disparities in the sex distributions of educational outcomes, with women opting out of STEM fields to a greater extent than men. Several factors probably play into this, the proposed chiefs among them being temperamental preferences for social/people-based disciplines, aversion to mathematics-heavy fields, social/societal conditioning and stereotypes (all influenced by gender roles and their interactions with sex differences), and uncertainty around what programming entails combined with lack of confidence in their own ability to perform (influenced by higher trait neuroticism on average). Since my experiment is an intervention study designed to increase mathematical, programming, and CT skills by way of early cognitive training, and since I will also collect data on confidence, preference for complex problems, and personality dimensions, I am thus well positioned to evaluate how aspects of personality and confidence interact with aptitude training to potentially increase performance in STEM-critical skills, primarily mathematics and CT (which is now starting to make its way into school curriculums). If we are interested in shifting this gender/sex-balance, this type of simple intervention might go some of the way towards alleviating early potential barriers diverting some women away from further pursuit of more technical studies and interests.

Open Science Practices (OS) & Research Data Management (RDM): A core idea of this project is for its methods and results to be openly available and reusable by both researchers and educators. To this end I will detail my RDM practices in the mandatory Data management plan (DMP) (**D5.1**) together with assistance from the KI research data office (RDO). Part of this effort will include uploading my datasets (**D2.1** & **D3.1**) to Zenodo together with my translated psychological instruments and the course material too, if agreed to by the original authors. Data will be

analysed with R and Python, since they are freely available under the GNU licence. Scripts, instructions for my datasets and how to obtain the course material in case open access is denied will be uploaded with the data to ensure the mandatory reproducibility. I will also upload my final manuscripts (**D5.3-4**). I will endeavour to publish my manuscript in an open peer-reviewed journal, and make my manuscript available on my ResearchGate profile. The Vektor WM training app is free to use by any interested teacher, barring server hosting limits on active accounts, but access to both teachers and researchers is controlled by professor Torkel Klingberg, who owns the IP. This is as open as possible given the inherent limitations. To comply with RDM best practices during the project, all paper-based data will be scanned and stored on our secured Lab servers, managed by KI, and the physical copies archived at our department. To cover the GDPR issues surrounding personal data only pseudonymised or aggregate datasets can be completely open access, but this together with my uploaded annotations will be enough to satisfy the FAIR requirements (further details will be included in the DMP). I will also consult dmp.ki.se, a KI tool that helps with creating, reviewing, and sharing data management plans that meet institutional and funder requirements. It is provided by the Digital Curation Centre (DCC).

1.3 *Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host*

Professor Torkel Klingberg is a senior researcher with over 20 years of experience as a PI, and is since 2013 a member of the Nobel Assembly. He has mentored 15 postdocs and 5 former PhD students to completion, and has a long track record of high quality publications in journals like PNAS, Nature Neuroscience, Developmental Cognitive Neuroscience, and Science, with over 28000 citations and an H-factor of 59, 7 articles cited >1000 times, 16 articles cited >500 times. He is an expert in the field of cognitive training and has published papers detailing the transfer effects from WM training to Mathematics, and how the brain is affected by this training. He has also developed the Vektor WM training app around which this project proposal is based. The Klingberg lab is small, hosting two current PhD students and the only postdoc currently being myself. This means that there will be ample time for the PI to oversee my project and to monitor progress by means of weekly meetings. My PI is committed to support me in accordance with the MSCA guidelines on supervision, and that they have read them. Progress updates will also be shared with the whole lab via lab meetings, where interesting papers or issues relating to other projects in the lab are discussed, also weekly. This gives me an opportunity to both learn new methods and concepts, but also to share the lessons from this project and my previous experience with other researchers in the lab continuously. The mandatory Career Development Plan (CDP) will be written together with KI according to an institutional template if this project is awarded, and this will be a deliverable in WP4 (**D4.1**).

Through this project I will gain significant research management skills in the form of project planning, budgeting, grant writing, managing school contacts, and through the KI MSCA Bootcamp in preparation for this application I have also become familiar with the EU Horizon Europe program, which I intend to use when searching for future funding for junior researchers. If agreed, I could also officially co-supervise a PhD student. The Klingberg lab has good knowledge of EEG, tACS and computational simulations and modelling approaches mainly using R and Python which is very useful to familiarise myself with for use in this and future projects. I bring my experiences of fMRI analysis and design, DCM analysis, Matlab software, qualitative research knowledge in education (particularly for computer-science) and knowledge of the interdisciplinary field of embodied cognition. I also bring networking opportunities since I have good relations with four professors from UEA in the UK, all working in different areas of research. I have already facilitated a seminar talk from one of them (professor Will Penny) in our institution, and hopefully more will follow.

In terms of training, KI offers several courses. Since I aim to improve my mentoring skills, I will enrol in the “Web course for supervisors of doctoral students” which is mandatory for supervisors of PhDs. This will then allow me to apply for a slot in the “introductory course in doctoral supervision” course. I will also attend the doctoral level course “Introduction to R”, since it is part of my open science plan discussed in 1.2. The Ki Career service also hosts courses, and I intend to attend the workshop “Psychological Safety - Implications for teaching and learning” and “Navigating Global and Diversity Equality Inclusion - Specific Challenges - (Particularly for Women’s Gender Gaps)” since teaching is a focus of both my project and my career plan. I will also attend the course “Academic productivity: Work smarter not harder/Work more effectively”. All these are key transferable skills.

1.4 *Quality and appropriateness of the researcher's professional experience, competences and skills*

I have a very eclectic background to date: medical science, genetics, psychology, stress, alcoholism, ageing, memory, cognition, self-image and sexuality, but my current line of research into programming really began with my doctoral thesis work, where I pioneered the fMRI study of the cognitive architecture underpinning successful programming learning in complete novices using an fMRI-paradigm invented by myself, giving us as clean a slate as possible from which to observe changes in brain and behaviour when learning computational thinking and what cortical areas are responsible for three hierarchical levels of processing. This proposal is exactly the kind of follow-up work enabled by what I did before, and I am one of few researchers uniquely positioned with one foot in cognitive neuroscience and one in computer science education to continue this research into the future. I came up with the concept of this project myself, and I hope that the writing of this proposal combined with my experience of running several large projects with human participants, and presenting my results at top international conferences like OHBM for brain imaging data twice, shows that I can competently design and organise a project of quality and disseminate the results. The greatest weakness of my current CV is that since my UK thesis is a monograph, I am lacking in first authored papers compared to most EU postdocs, but I have recently submitted my manuscript based on the key findings from the thesis to Trends in Neuroscience and Education that is currently under review. What I do have however, is the grit to persevere through difficulties, and a proven track record of forging collaborative bonds with several established researchers who I am actively collaborating with still, and the experience of running administratively complex experimental procedures involving participants of various ages, from minors all the way to retirees using many varied methodologies. I have already performed a two-week pilot study based on this proposal, so I am familiar with how to perform the aptitude testing on the participating children successfully, and I have been able to build up a network of interested teachers and headmasters from eleven schools so far willing to participate. The pilot demonstrated that the course material resulted in an increase of $6.4 \pm 3.2\%$ score on the programming test per day of study, and that I can comfortably assess 7 children in a day of work (see 3.1). I have experience of communicating the potential of this field of research to educators and software professionals from my panel discussion at the Trippel Helix conference, and since starting my postdoc I am actively attempting to take on more roles of responsibility wherever I can to contribute to stakeholders in my circle of influence, the latest examples being: Volunteering via the "Skype a scientist" science outreach charity, asking to help out with teaching again at the Psychology and Neuroscience departments at KI, intending to co-supervise a PhD student, and serving on the board of my housing association. I have also read the European Charter for Researchers, and will follow it in order to represent the best of conduct in all interactions.

2. **Impact** #IMP-ACT-IA@#

2.1 *Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to his/her skills development*

The most important opportunity this MSCA fellowship would offer is the opportunity to produce two manuscripts (D5.3-4) of high quality that would directly address the biggest need in my CV that I will need to address in order to increase my chances of securing the future funding to eventually establish myself as an independent researcher. This project aims to address three different objectives, so the potential to produce several manuscripts is there. If awarded, this fellowship will cement the fact for future employers and grant bodies that I am capable of planning original research, securing funding, and administering a project to completion.

I intend to use my experience of the MSCA fellowship to apply for an ERC Starting Grant, allowing me to establish a lab dedicated to studying the intersection of programming, cognition, and learning in Sweden, continuing to capitalise on the network of participating schools established in this project. This will continuously benefit me and any future employers or collaborators who want to involve more teaching professionals in their research. Involving teachers might also contribute to future R&I output better tailored to the interests of this group of stakeholders. This project also has a high probability of leading to further projects, since it is breaking new ground and looking at several aspects of programming learning. I have already written another grant proposal together with my PI and with his guidance, this fellowship will allow me to further deepen my transferable skills in the project planning, administration, and grant writing department. The courses offered by KI detailed in 1.3 are also aimed at improving transferable skills. In terms of academic positions, short term I will aim towards applying for "Biträdande lektor" (associate lecturer) after completing a few years as postdoc, with the long-term goal of eventually going for a tenure track career.

2.2 Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities #@COM-DIS-VIS-CDV@#

The final version of my dissemination, communication and exploitation plan will be written with input from my PI and the KI research support office. In accordance with the European Charter for Researchers, I will do my utmost to ensure that the results of my research are disseminated and exploited to best effect, since MSCA is publicly funded. It will include expectations on From the end of September 2023, employees at KI are able to use KI RIMS (Research Information Management System), a new database for projects with the goal of collecting, improving and make the information about KI's research available in a centralised and easily searchable format, allowing for interested parties to find relevant research outputs and to achieve increased dissemination of information about ongoing research without increased administration or for KI to help spread awareness to various channels from this database. I will make full use of this resource for my project.

Dissemination activities: The main message of my dissemination activities will be the results as they stand at the time of each activity, the objectives of the project (see 1.1), and the expected impacts (see 2.3).

Activity	Target Audience	When	Where	Key Indicators
<i>Two Manuscripts for publication (1:O1 & O2, 2:O3)</i>	<i>Researchers (Psychology, Education and Computer Science), Education Professionals, Policy Makers</i>	<i>Match Month 24 (D5.3-4)</i>	<i>“Education and Information Technologies”, “Transactions on Computing Education” or “Computer Science Education”</i>	<i>H-Indexes: 76, 47, 43.</i>
<i>Attend a scientific Conference.</i>	<i>Researchers (Psychology, Education and Computer Science), Education Professionals, Policy Makers</i>	<i>April Month 13 (M5.2)</i>	<i>CHI (Conference on Human Factors in Computing Systems), ICER (International Computing Education Research Workshop) or ITiCSE (Annual Conference on Innovation and Technology in Computer Science Education). (Locations change yearly)</i>	<i>Thousands of attendees. Average Downloads: 933, 491, 388. Average citations: 23, 14, 8.</i>
<i>Annual KI StratNeuro Retreat*</i>	<i>Neuroscience Researchers from KI, Umeå University, and the Royal Institute of Technology.</i>	<i>June Month 15 (M5.3)</i>	<i>Stockholm.</i>	<i>150+ participants</i>
<i>Pint of science, Sweden**</i>	<i>Scientists, students & the general public.</i>	<i>May Month 2 or 14 (M5.1)</i>	<i>Stockholm.</i>	<i>885 Followers on Facebook</i>
<i>European Researchers' Night (MSCA)</i>	<i>The general public. (Dissemination & Communication)</i>	<i>September Month 6 and/or 18. (M6.1)</i>	<i>Stockholm.</i>	<i>>1.5 million visitors annually.</i>

Table 1 - Dissemination activities. (Multiple options presented in order of preference.)’

Project COGITO

* The StratNeuro initiative funds an annual conference retreat for researchers in Neuroscience free to attend if accepted to present. StratNeuro involves KI, Umeå University, and the Royal Institute of Technology.

** Pint of science, is originally a British concept where short scientific talks are held in a pub.

Communication activities: The KI StratNeuro retreat (**M5.3**) contains a significant element of communication as well, namely the opportunity for finding future collaborative partners outside my own department. I will also offer to present my project at the departmental Psychology seminar series.

Stockholm participates in the European Researchers' Night (**M6.1**), so I intend to apply to participate representing the Neuroscience department at KI. This would allow me to communicate with the general public about what we do in the lab, how it is to work in science, and to raise awareness of the R&I questions and objectives touched on by this project and how they impact everyday life (see 2.3).

During this project I will be interacting and visiting several schools, and I will hold feedback sessions after each cohort to give the teachers a chance to ask questions and provide their thoughts on the experiment and future possibilities or interests (**D6.1-2**).

Since I attended the KI MSCA Bootcamp in preparation for writing this proposal, I will return as an alumnus to support future applicants to the MSCA postdoc fellowship if I am awarded (**M6.2**).

I will advertise the start of this project, the results of the first cohort, and the final Research outputs on my private account on x.com(Twitter) (**D6.3**). (I will also most likely advertise other milestones, like conference presentations in addition to this.) This will be part of an effort to expand my interactions with accounts dedicated to science communication and will hopefully lead to a future where I can use x.com to advertise my research and results to a diverse audience. Currently my account has 336 followers.

Exploitation activities: If WM training improves programming learning and/or computational thinking skills, then we have 300 children already positively affected. All 600 will also have trained programming in any case, and hopefully benefit from this as well. Teachers and headmasters from the recruited schools will be free to continue to use the material from this project, I will encourage them to do so in all my interactions and all my results will be published as Mandatory open access without embargos. (KI has a prepaid APC contract for open publishing with several journals, like PLOS, Frontiers, Springer, Elsevier and Taylor & Francis so I will consider using these for my open access publishing to reduce cost.) I will also volunteer to present my findings or lecture to any of the involved schools to further engage with the professional education community and ensure pertinent results reach policymakers in the form of principals.

The results from this project have the potential to benefit future studies on programming and WM, lead to new scientific questions, and it has the possibility to result in a standardisation of WM training and ScratchJr teaching in primary schools. To further increase exploitation, I will inform the authors of the CAL course of any pertinent result or uses of their course material so they can update their website and perhaps collaborate with me in the future. This would be exactly the type of interdisciplinary collaboration between education- and neuroscientists I am looking for.

Exploitation potentials in summary: 1. Further research, 2. Collaborations, 3. standardisation, 4. engagement with end users & policymakers.

IP strategy: All my translated materials will be made freely available to the extent that the original authors agree. When it comes to all other IP, including the data and publishable material, they fall under Professor's privilege in Sweden, meaning that I will hold all copyrights to these materials and associated patent rights. All these rights can also be freely transferred. If needed KI Research Support Office offers help with all IP related issues. KI Innovations AB (KIAB) is also an initiative from the university that can help researchers with developing their ideas into actionable products or services to maximise impact potential. I have read, and will comply with the MSCA grant agreement rules for IP.

2.3. The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts

Scientific Impacts: As discussed in 1.2, this project will use open-source solutions where possible and provide all material needed for reuse by other scientists. Short term, I will produce two peer reviewed publications of high quality, aiming for journals where papers are downloaded >900 times on average (see 2.2).

Long-term, this project will begin to paint a clearer picture of what uniquely constitutes computational thinking, and how this relates to working memory and broader cognitive processes. If working memory training in conjunction with early exposure to programming concepts leads to improvements in computational thinking, then we have both further confirmed the link between programming and VSWM, and found a way to improve children's general cognitive skill development. This would then open up the question of whether this improvement has lasting knock-on effects translating into better performance throughout their educational life. Further longitudinal studies of children with and without this early intervention could then be designed. The Vektor app allows effective monitoring of usage, allowing for easily measured evidence of adoption.

Societal Impacts (Long term): If future schooling is to implement a mixture of common and individualised educational material, a better understanding of the interplay between cognitive and non-cognitive attributes will be of paramount importance in tailoring the learning experience to each individual child, this project is a step in that direction. I will impact 600 children during this two-year period, with the potential for more in future cohorts of further testing. ~110,000 children enter primary school every year in Sweden alone, so if this project is successful and implemented nationally the impact would be significant. Wider understanding of what programming entails as fostered by the course material could also create a more resilient, inclusive, and democratic European society by strengthening STEM awareness, while also fostering key digital skills (part of the value-chain of accelerating the digital transition).

Economic Impacts: In the medium to long term, this project might provide evidence that early cognitive-mathematical training intervention paired with early exposure to programming not only improves programming learning outcomes but might also transfer over to broader computational thinking skills. If this is the case, then pairing these two activities in primary school can be cheaply and easily implemented to improve children's STEM prospects. Since the individual and societal benefits of investment in early learning are also vastly more cost effective than later investments (Elango, 2015) it makes sense to prioritise early mathematical learning considering the central role mathematical skills seem to occupy both in relation to programming and general achievement. Given that the ICT sector in the EU was over €718 billion in 2021 (5.5 % of the EU's Gross value added), fostering IT skills is of high importance.

#\$COM-DIS-VIS-CDV\$# #IMP-ACT-IA\$#

3. Quality and Efficiency of the Implementation #@QUA-LIT-QL@# #@WRK-PLA-WP@#

#@CON-SOR-CS@# #@PRJ-MGT-PM@#

3.1 Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages

As detailed in 1.2, this project is divided into two cohorts of data collection (**WP2-3**) with preparatory work running concurrently in **WP1**. Since I am using schools, the schedule must conform to the school year with breaks accounted for. The structure of the project is roughly: 1. Translate and prepare all materials and instruments (**T1.3**). Then, for each cohort: 2. Instruct teachers and collect consent forms (**T1.1**). 3. Teachers collect instruments and train WM with children. 4. Teachers administer the CAL course. 5. I administer the CSA test of programming for two groups of 150 children in a staggered timing (**T2-3**), allowing all data collection for each cohort in under a month total time with assistance from 2 assistants. 6. data organisation and analysis (**T2.3 & T3.3**). Most of my work will be spent in administration and recruitment, running continuously when not collecting and analysing data at the end of each cohort.

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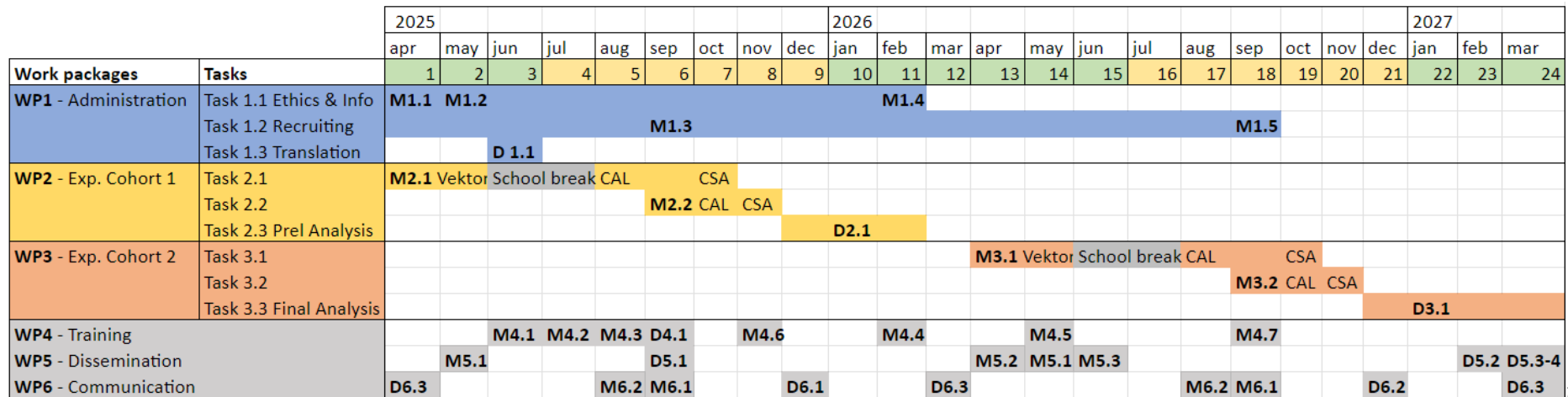


Fig.1 - Gantt

Chart.

WP 1 Administration	Task 1.1	Ethics Permit & Information meetings with teacher cohorts	M1.1 Ethics approval	M1.2 Information meeting with teachers, cohort 1	M1.4 Information meeting with teachers, cohort 2
	Task 1.2	Recruit RA:s and Teachers	M1.3 RA:s recruited, cohort 1	M1.5 RA:s recruited, cohort 2	
	Task 1.3	Translate Instruments and course material	D1.1 Translated instruments & CAL course material		
WP 2 Cohort 1	Task 2.1	Group 1: vektort training, CAL course, TechCheck & CSA evaluation	M2.1 Collect informed consent		
	Task 2.2	Group 2: CAL course, TechCheck & CSA evaluation	M2.2 Collect informed consent		
	Task 2.3	Preliminary Data Analysis	D2.1 Preliminary dataset		
WP3 Cohort 2	Task 3.1	Group 1: vektort training, CAL course, TechCheck & CSA evaluation	M3.1 Collect informed consent		
	Task 3.2	Group 2: CAL course, TechCheck & CSA evaluation	M3.2 Collect informed consent		
	Task 3.3	Final Data Analysis	D3.1 Final dataset		
WP 4 Training	M4.1	Academic productivity: Work smarter not harder/Work more effectively	M4.5 Web course for supervisors of doctoral students		
	M4.2	KI agenda 2030 SDG course	M4.6 introductory course in doctoral supervision		
	M4.3	Psychological Safety - Implications for teaching and learning	M4.7 Navigating Global and Diversity Equality Inclusion - Specific Challenges - (Particularly for Women's Gender Gaps)		
	M4.4	Introduction to R	D4.1 Career development plan (mandatory)		
WP5 Dissemination	M5.1	Pint of Science Pub Talks	D5.2 Dissemination, Communication and Exploitation Plan (mandatory)		
	M5.2	Scientific Conference	D5.3 Written Manuscript 1		
	M5.3	KI StratNeuro Retreat	D5.4 Written Manuscript 2		
	D5.1	Data Management Plan (mandatory)			
WP6 Communication	M6.1	European Researchers' Night	D6.2 Teacher feedback session 2		
	M6.2	KI MSCA Bootcamp Alumni	D6.3 x.com(Twitter) Dissemination & Communication Announcements		
	D6.1	Teacher feedback session 1			

Table 2 -

Work Plan Table.

The research progress, training and career planning will be monitored together with my PI during weekly meetings. **Administrative risk (low):** Collecting psychological profiles of children requires ethical approval (**M1.1**). Ethics already applied for 31/07/2023, Ref. 2024-04879-01. **Financial risk (low):** Project requires very little material. My department head has already signed off on the co-financing costs of overhead coverage. **Operational risks (medium):** Recruiting and coordinating with multiple schools can be difficult, threatening our sample size. To mitigate this, I have already performed a successful pilot, and recruited 11 schools, >20 teachers with ~20 children each, covering my need for 300 participants per cohort. My PI will also advise since he has experience of conducting experiments in schools. The second risk is the large data collection work load, 1-hour evaluations for each of the 600 children. To mitigate this, only 150 children will participate at a time and two students or RAs will be hired to help with the collections, so each of the 4 collection points should take maximum of two weeks if each person processes 6 children per day, which is less than what I could comfortably achieve during my pilot of the material. **Scientific risk (medium):** Since the hypothesis is novel, there is a risk that **O1** shows no transfer effect. In this case, the experimental design allows us to still analyse the interactions of within-subject factors between training and mathematics metrics, the personality measurements, programing aptitude and computational thinking score. This gives us a good foundation from which to describe the cognitive and non-cognitive processes and how they impact the training progression and outcome measures even if the intervention fails to produce any between group effects.

3.2 *Quality and capacity of the host institutions and participating organisations, including hosting arrangements*

At KI I have been given a dedicated office space, including storage space, stationery supplies and IT materiel. I have been integrated into the Klingberg Lab and actively take part in journal clubs, progress reports from team members and seminars hosted by the departments of Neuroscience and Psychology. My colleagues have quite technical backgrounds, so the opportunity to learn more statistics, modelling, and coding languages is high. Meetings with my PI are held at least weekly. My PI and the department are able to sponsor this project financially if necessary.

The department hosts annual Christmas retreats where presentations of research in the department are held to foster integration amongst the labs. KI has the MSCA Bootcamp mentioned in 2.2. I have also joined KIPA, the KI postdoc association, that organises social events on a regular basis. The StratNeuro initiative, also mentioned in 2.2, allows for contact with other universities in sweden doing neuroscience work, and they host seminars and an annual conference retreat. KI has a career service that hosts training and 1 on 1 sessions to discuss career development free of charge. The KI library also hosts transferable skill courses like the open science workshop, and provides access to publications and other databases. KI has a research support office, also providing seminars and courses. The grant office sends out regular newsletters with funding opportunities and gives access to a PIVOT database. As mentioned in 1.2, I will also have access to dmp.ki.se, a tool that helps with creating, reviewing, and sharing data management plans that meet institutional and funder requirements, provided by the Digital Curation Centre (DCC). KI has received the EURAXESS HR Excellence in Research award, Awarded on 08-06-2020.

#§CON-SOR-CS§# #§PRJ-MGT-PM§# #§QUA-LIT-QL§# #§WRK-PLA-WP§#

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Part B-2**4. CV of the researcher****Andreas Lidström (PhD) | Cognitive Neuropsychologist**

Location: Stockholm, Sweden

Telephone: +46(0)739977651

Email: antecave@gmail.comandreas.lidstrom.2@ki.se<https://www.researchgate.net/profile/Andreas-Lidstroem/research><https://orcid.org/0009-0008-7372-8541>**Research Profile & History**

My primary areas of expertise are fMRI analysis and design & the planning and running of brain imaging experiments. I am familiar with a variety of statistical modelling techniques, such as SEM, GLM, FIR, ICA, and DCM, and can use several of the most common software packages. My PhD research project was an interdisciplinary collaboration between researchers from the fields of neuropsychology and computer-science education into the neural correlates of learning computer programming, giving me experience of both qualitative and quantitative research. After that I have studied memory and ageing using interviews, EEG and sense of smell. Before my PhD, I have also worked with DNA sequencing, stress metrics, and alcohol withdrawal in rats, before finally turning to cognitive neuroscience, and studies of emotional priming effects, burnout, and gender dysphoria. The subjects may be varied, but my primary interests lie in the intersection of behaviour, personality, and biology, such as my current field of 'mind, brain and education'.

My background is somewhat unusual and interdisciplinary, since I began my studies by obtaining a BSc and MSc in medical science before deciding I wanted to work with fMRI in cognitive neuroscience, which eventually led to a PhD in Psychology from UEA in the UK, and I very much wanted to stay living there in the university town of Norwich. Unfortunately I got caught up in the hiring freeze at UEA after covid struck at the end of my PhD, leading to a stop in my teaching duties, and I subsequently spent over a year looking for postdoc positions at other universities in the UK or abroad matching the research profile from my PhD, while living solely off of my investments in Bitcoin. I eventually started working on two different projects concurrently via two very short-term temporary research assistant positions under professors I had come to know during my time as a PhD student, before finally being offered a Research Associate position at UEA from a third professor, but for 8 months only. The continuing issue being lack of funding at the school rather than lack of enthusiasm from professors. This at least gave me time to evaluate my options, and I finally made the hard decision to start fresh by accepting my current postdoc position, moving back to Sweden and making an attempt to continue my research here by building on my experiences gained from my time in the UK. I see the MSCA PF as a great opportunity to help me solidify my profile as an emerging leader in the field of Programming Cognition, allowing me to capitalise on the novel findings from my PhD by testing the limits of the real world potential of early WM training, and providing me a good platform of high quality publications, from which to apply for further startup grant-money to continue my research career. I see this project paired with my PI as the perfect vehicle for this ambition (B1.1-4).

Professional Experience

04/12/2023 - Present (ends 03/12/2025)
Postdoc

Karolinska University, Department of Neuroscience

Postdoc position, studying the effects of personality differences and working memory training on programming learning and computational thinking, and applying DCM to longitudinal data of working memory training together

with Professor Torkel Klingberg.

Key responsibilities

- Planning and conducting Research
- Applying for Grants
- Advising PhD students
- Presenting at Journal clubs

26/06/2023 - 24/11/2023

**University of East Anglia, School of Psychology
Research Associate**

8 month Research Associate position, Project organisation and data collection (fMRI, EEG, Genotyping, olfactory performance and structured interviews) in healthy ageing project, assessing memory and pre-clinical dementia as a function of APOE4 status together with Professor Louis Renoult.

Key responsibilities

- Organisation of workflow and participant processing.
- Data collection and quality assurance

01/05/2022 – 01/06/2023

**University of East Anglia, School of Psychology
Research Assistant**

Part-time Methodological consultant on DCM analysis in a fNIRS project investigating connectivity changes relevant to cognitive functioning in infants affected by malnutrition from India, and collected EEG data for a project to investigate representational processing during quiet wakefulness together with Professors John Spencer and William Penny.

Key responsibilities

- Help establish institutional DCM knowledge in a new research group for future projects.
- Design and run DCM analyses on the fNIRS data set and produce and interpret the results.
- Collecting EEG data

01/10/2017 – 30/09/2021

**University of East Anglia, School of Psychology
Staff Research Associate & registered PhD Student**

Research into the neural correlates of learning to program computers in novice students of programming, resulting in the thesis "Towards a Neuroscience of Computer Programming & Education". I designed, ran, and analysed the data for the fMRI study using GLM-, FIR-, SEM- and DCM models to answer fundamental questions regarding the nature of programming knowledge and acquisition, including the role of confidence and motivation.

Key responsibilities

- Experimental design, data acquisition and analysis.
- Results presentation in the form of posters, seminars, and thesis writing.
- Liaising with project collaborators.
- Research subject recruitment and processing.
- Associate tutor, "Research Design and Analysis 1" undergrad statistics module.

01/09/2016 – 30/06/2017

**Karolinska Institutet, Institution of Clinical Neuroscience
Research Assistant**

Preparatory work for my PhD project (originally planned to commence in Sweden), researching methodologies, conferencing with our collaborators, and mentoring a master's student.

01/10/2013 – 30/11/2014

**Karolinska Institutet, Department of Women's and Children's
Health and Neurology Clinic
Research Assistant**

Behavioural measurements and fMRI acquisition and analysis of Resting-State functional connectivity in homosexual and transsexual populations using seed-region and independent component analysis. fMRI and EOG analysis of burnout patients pre- and post-treatment.

Positions of Trust, Project Work & Volunteering

June 2024 - Present

**Member on the Board of Directors and signatory for BRF Snipan,
condominium housing association**

January 2013 – June 2013

**Karolinska Institutet, Department of Neuroscience, Division of Psychiatry
Master's thesis work**

Final project course involving a preclinical experiment of antidepressant medication for treatment of alcohol use disorder, utilizing the 'Intermittent-access 20% ethanol 2-bottle-choice model'. Title: "The Effects of LY2607540, a Positive Allosteric Modulator of the mGlu2Receptor, on Voluntary Ethanol Intake in Rats".

October 2012 – January 2013

**Karolinska Institutet, Department of Clinical Neuroscience
Elective Project work**

I performed a socio-cognitive behavioural study on healthy volunteers using a language task to evaluate how self-perception and free choices influence cognitive performance. Title: "Free Choices and the Unconscious Effects of Self-Perception".

January 2011 – June 2011

**University of Newcastle, Neurocardiology Laboratory, School of
Biomedical Sciences
Bachelor's thesis work**

Final project course as an exchange student in Australia, resulting in the thesis titled "Relations between cortical and respiratory arousal in rats" where I was evaluating the reliability of whole-body plethysmography as compared to EEG and ECG in stress research.

**2009 & 2010 Karolinska Institutet, Department of Medical Biochemistry and Biophysics,
Division of Medical Inflammation Research
Volunteer work**

Volunteered during summer breaks for lab experience. I performed a genotyping study on rat spleens using TaqMan QPCR to look for SNP variations that could be linked to Rheumatoid Arthritis. Next year I performed a sequencing study on the same samples using dye terminator sequencing. I sequenced 25000 bases spanning ten genes in two different rat strains.

Education and Qualifications

- **Doctor of Philosophy** – University of East Anglia, School of Psychology
– VIVA passed 09/12/2021, formally awarded 21/07/2022
- **Master of Medical Science** – Karolinska Institutet – 20/06/2013
- **Laboratory Animal Science, FELASA C** – Karolinska Institutet – 17/02/2012
- **Bachelor of Medical science** – Karolinska Institutet – 12/07/2011
- **Certificate in Advanced English, Grade A** – University of Cambridge – 05/08/2008

Peer-Reviewed Publications

Menghi, N., Melega, G., **Lidström, A.**, Renault, L., & Penny, W. Delayed Emergence of EEG-based Task-relevant Representations - Manuscript Submitted, Neurobiology of Learning and Memory, NLM-24-114.

- A follow-up study aimed at assessing the effects of periods of offline wakefulness on generalization of rules and memorization, with implications for studies of effective learning. I was responsible for collecting the EEG data.

Lidström, A. & Bengtsson, S. Broca's language area is central to teenage novices' computer programming learning - Manuscript Submitted, Trends in Neuroscience and Education, TINE-D-24-00039.

- Manuscript focusing on the core findings from my PhD thesis work, detailed below.

Lidström, A. (2021) Towards a Neuroscience of Computer Programming & Education, Monograph, UEA Digital Repository.

- The first comprehensive examination of the foundational cognitive machinery underpinning programming learning using fMRI and dynamic causal modelling of connectivity. The roles of confidence, Need for Cognition and several educational attainment measures were also investigated with regards to their interactions with the learning outcomes.

Feusner, J. D., **Lidström, A.**, Moody, T. D., Dhejne, C., Bookheimer, S. Y., & Savic, I. (2017). Intrinsic network connectivity and own body perception in gender dysphoria. Brain imaging and behavior, 11, 964-976.

- Well-cited study investigating brain network connectivity patterns involved in own body perception in the context of self in gender dysphoria. Foundational research into what may represent the neurobiological correlates to the subjective disconnection between perception of body and self-identification. I ran the fMRI experiment and performed the analysis of the functional brain imaging data after processing it with manual independent component analysis.

Nalivaiko, E., Bondarenko, E., **Lidström, A.**, & Barry, R. J. (2012). Respiratory component of the orienting reflex: a novel sensitive index of sensory-induced arousal in rats. *Frontiers in physiology*, 2, 114.

- Validated an experimental setup for measuring stress responses in freely moving rats without the need for invasive EEG or ECG measurements, improving ecological validity of stress experiments and reducing animal suffering. I ran the behavioural part of the experiment and calculated the inter-modality response differences.

Other Publications

Learning concepts in computer programming, 2019 - poster presentation, International Symposium on Motivational and Cognitive Control

Adopting Relevant Concepts in Computer Programming, 2019 - poster presentation OHBM Rome

Altered network connectivity and own body perception in gender dysphoria, 2015 - poster presentation

Rs-fMRI shows Disconnections in Own Body Perception Areas in Female to Male Transsexuals, 2014 - poster presentation OHBM Hamburg

Grants and Awards

First place at Sweden's annual science fair, National Technology Museum, 2008. - Small grant for programmable robot project, showcasing voice control and computer-vision with applications for people living with disabilities.

Invited Presentations

Lecture - Danderyd high school, Sweden 21/11/2017 - Presented a lecture on cognitive neuroscience and brain imaging to students and teachers from one of the schools from where participants for project HOPE were recruited.

Tripple Helix conference 08/09/2017 - Engaged with key stakeholders from the private and public sector by spreading awareness of project HOPE via participation in a discussion panel at the Trippel Helix Conference on Computational Thinking and Digital Competences in Primary and Secondary Education, Stockholm, September 8, 2017. Organised by Swedsoft, a non-profit organisation working to promote competitive software engineering in Sweden.

Supervising and Mentoring

Teaching/grading - Taught the practical workshops for the course 'Research Design and Analysis 1' to first year undergraduates at the University of East Anglia and graded final exam project reports. 2019

Master student - Supervised Masters student Khadija Osman through final year project work. 2017

5. Capacity of the Participating Organisation(s)

5.1 Template table: Overview of Participating Organisations

Organisation role	PIC	Legal Entity Short Name	Academic organisation (Y/N)	Country	Name of Supervisor
Beneficiary	999978530	KI	Y	Sweden	Torkel Klingberg

5.2 Template table: Capacity of the Participating Organisations

Choose one of: ☒ <i>Beneficiary (compulsory)</i>	
Karolinska Institutet, KI, Sweden	
Medical University	
Role and profile of supervisor	Professor Torkel Klingberg's research is aimed at better understanding the neural basis for cognitive development during childhood, including training-induced brain plasticity. He is primarily interested in the psychological functions of working memory, attention, and reasoning, and how this relates to academic abilities such as mathematics and reading in addition to the development of computerised methods for training and evaluation of working memory and mathematical abilities in children. Beside active supervision of the planning and execution of the proposed project to ensure feasibility of planned work and deliverables, Torkel will also assist me by offering his extensive experience of over 20 years of project management-, research dissemination-, supervision-, and grant writing-skills as a PI to further my own future academic prospects via the improvement of these key transferable skills.
Key research facilities, Infrastructure and Equipment	The Klingberg Lab at the department of Neuroscience at KI will supply me with office space for working, access to stationery as necessary, secure digital and physical storage of research materials and data. Core departments of KI will assist me administratively with institutional support-organs for policy compliance and training courses in the realms of IP management, data management, finance, ethics and EU regulatory frameworks. For a detailed list of planned training activities please refer to WP4 (Training) and WP5 (Dissemination) in section 3.1. KI research support service also hosts an 8 session bootcamp for MSCA applicants, which I have attended.
Previous and current involvement in EU-funded research and training programmes/actions/projects	Since 2014 KI has successfully hosted 61 MSCA postdocs and projects within the Horizon 2020 and Horizon Europe framework programs, 12 of which were awarded in 2023. KI is also one of the eight members of NeurotechEU, an alliance of universities which aims to build a trans-European network of excellence in brain research and technologies by 2040. NeurotechEU is a part of the EU's European Universities initiative.

6. Additional ethics information

Psychological Instruments:

The Vektor training program is an application developed by Klingberg et al., freely available on the web, designed to be used with tablets by children. It includes both pre- and post-assessments of mathematics and has continuously monitored training questions to track learning over time (rate of cognitive improvement), divided between NVR, VSWM, and number- line tasks (Judd and Klingberg, 2021).

The Need For Cognition scale is a six item scale designed to measure attitudes towards learning, specifically preference for or aversion to challenging intellectual problems (Cohen, Stotland, & Wolfe, 1955).

The Rosenbergs Self-Esteem scale is a widely used 10 item scale designed to measure self- esteem (Rosenberg, M 1965), a modified version adapted for children under 12 years of age will be used (Wood C et al., 2021). The Pictorial Personality Traits Questionnaire for Children (PPTQ-C) is a 15 item scale designed for children between 7 and 13 years of age, incorporating graphical illustrations of behaviors representing the two poles for each question item (Maćkiewicz, M., & Cieciuc 2016).

The Coding As Language' (CAL) introductory programming course is a full 25 day curriculum, with accompanying assessment materials designed to introduce children to programming with a custom textless version of the Scratch block-programming language called ScratchJr. ScratchJr is one of the most popular tools for teaching programming concepts to young children, and as of a 2018 review, 2% of Swedes have used ScratchJr, (potentially up to about one out of three children in the 5-9 age range) (Bers, 2018). The course is designed to teach broader computational thinking concepts alongside the practical work, to anchor concepts to positive attitudes and values, and to convey the potential for creative applications of programming, like story-telling (Bers M. U. et al., 2023).

The Coding Stages Assessment (CSA) is the assessment material designed to measure several dimensions of programming aptitude at the end of the CAL course. These are divided into five "coding stages" (Emergent, Coding and Decoding, Fluency, New Knowledge, and Purposefulness) and can be used to quantify the level of understanding achieved (De Ruiter et al., 2022).

The TechCheck is an additional assessment developed to be administered at the end of the CAL course. It is a 15 items long multiple-choice instrument that is designed to probe six domains of computational thinking (algorithms, modularity, control structures, representation, hardware/software, and debugging) (Relkin, E et al., 2020)

Analysis:

This experiment is a controlled intervention trial in an ecologically valid setting (normal school classrooms). It will involve individual personality survey administrations, both in person or over the internet. The statistical analysis will principally involve regression modeling of our outcome measures, and structural equations modeling. Linking back to the aims of this study, to identify the learning-relevant cognitive components related to computational thinking we will correlate outcome and improvement rate for WM, NVR and number line training in the Vektor app with the coding stages achieved on the CSA and score on the TechCheck. Likewise, to Identify how personality traits and motivational factors relate to computational thinking and programming learning we will correlate the big five personality scores, self-esteem scores and NFC scores with the same learning outcomes, as described above.

Finally, to Investigate the hypothesis that training in either mathematics, visuo-spatial working memory (VSWM), or non-verbal reasoning (NVR) improves computational thinking and the learning of programming we will compare our training group to the controls. We will also model this relationship with our learning outcome variables while compensating for the personality measures to evaluate the relative contributions of training and cognitive skills contra non-cognitive attributes through regression and Structural Equations Modeling (SEM).

7. Additional information on security screening

Not applicable.

8. Environmental considerations in light of the MSCA Green Charter

Karolinska Institutet offers a two module online course designed to familiarise employees and students with the Agenda 2030 Sustainable Development Goals, which I will complete as part of my training (Work Package 4) to develop my awareness of environmental sustainability.

As part of the wider school digitisation initiative, this project will make use of tablets (owned by participating schools or lent) that are reusable for multiple purposes and consecutive years of students. Paper consent forms and handouts of the psychological instruments will only be used to ensure data integrity and for archival purposes at the host institution. As an example of good practice, the Klingberg Lab has made available 150 lendable tablets to schools participating in previous projects, ensuring maximum participation of interested schools without their own individual tablets for their children, while also reducing environmental impact by sharing resources.

The local traffic net is well developed in Sweden and environmentally conscious in its implementation. The Stockholm local trains rely on renewable energy sources for its electricity (solar-, wind- or hydropower), and since 2018 all local buses run on renewables in the form of biogas, ethanol, biodiesel or electricity. I will employ only local traffic, preferably trains when visiting the schools within commuting distance for the evaluations of the students. For more remotely located schools, or in the interest of scheduling, teleconferencing via Zoom (Service provided by Karolinska Institutet) will be used for the evaluations. I will also use Zoom to organise the information meetings with the participating teachers for optimal efficiency.

The Swedish power grid is primarily hydropower at ~30%, nuclear at ~20%, wind and other variable renewables at ~20%, with the remainder of the base load being primarily made up of oil. With the continued expansion of wind power and the planned construction of new modern nuclear plants to replace the current reliance on oil in the south, where hydropower is out of effective transfer range, Sweden is well positioned to meet future demand for power while still further reducing its already low carbon footprint. Investment in Swedish R&I initiatives should thus be favoured from an environmental consideration.

9. Required for Global Fellowships only: Letter(s) of commitment from associated partners (hosting the outgoing phase)

Not applicable.



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